S Srikanth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.44
Narayana PU college	Class XII (KSEAB)	2020	94.33%
Holy Spirit School	Class X (ICSE)	2018	84.20%

RELEVANT COURSES

1. Aerospace Engineering 4. Mechanics of Composites
2. Structural Mechanics 5. FEM (audit)
3. Machine Learning 6. High speed aerodynamics

INTERNSHIP AND APPRENTICESHIP

Apprenticeship training at HAL engine division, Bengaluru | (3 months)

- Worked at the Methods department, modelling and drafting Shakti Engine's fuel valve, GTRE gear casing and AMAGB components.
- Trained on basic GD and T, some of the manufactuing techniques used to manufacture a aircraft engine component.
- Drafted the schematics of Nitriding furnace in Heat treatment department.

Internship Training | Ace Micromatic Group & Associated Units, Peenya

Manufacturing of CNC Components

- **Gained practical exposure to diverse CNC machinery** (Turning Centers, VMCs, HMCs, Grinders) and specialized gear manufacturing processes, including hobbing, skiving, and grinding at Pragati Transmission.
- **Studied end-to-end shop-floor workflows** across production lines, analyzing automated tooling setups, machine layouts, and quality inspection protocols.

SKILLS

CAD / 3D Modeling: SolidWorks, CATIA, Siemens NX (UG NX)

CAE & Simulation: ANSYS Workbench, ANSYS Fluent, Abaqus, SolidWorks Simulation

Programming: Python, Java (Basic)

Software & Tools: Microsoft 365 (Excel, Word, PowerPoint)

Languages Known: English, Kannada, Tamil, Hindi

SCHOLASTIC ACHIEVEMENTS

Awarded the HAL (Hindustan Aeronautics Limited) Engine Division Scholarship (2020) grant of **Rs. 1 Lakh** for academic excellence in Pre-University examinations -- one of **10 recipients selected division-wide** among eligible employee dependents.

PROJECTS

PROJECT : Process-Induced Deformation & Residual Stresses in Hybrid Composite Laminates - In collaboration with DRDO ADA, NAL:

- Engineered a custom six-instance multi-shell FE architecture in Abaqus for a 36-ply hybrid CFRP/GFRP laminate, coding UMAT and UMATHT subroutines from scratch to resolve ply-by-ply thermal cure gradients beyond its native

20-point thermal integration limit.
- Benchmarked four matrix constitutive models -- mCHILE, Elastic, PD, and Viscoelastic -- against an analytical CLPT framework, isolating the viscoelastic formulation at 7.413 mm as the closest numerical predictor against a 9.15 mm experimental benchmark.
- Narrowed the process-induced warpage prediction gap to **1.74 mm**, nearly **2x tighter** than the baseline mCHILE model's 3.27 mm gap, by comparing numerical-analytical solutions across all four constitutive formulations under purely material-driven assumptions.
- Diagnosed the residual **1.74 mm** gap as a missing tool-part interaction effect -- locked-in shear stress released during demolding -- and authored a first-author paper submitted to the 6th International Structural Integrity Conference & Exhibition (SICE) 2026, under review.

PROJECT: Powered-Lift STOL Aircraft Design -- Inter IIT Tech Meet 14.0, Team LAT Aerospace (8-member team)

Solo contributions:

- **Designed and 3D-printed** a modular wind-tunnel test fixture in SolidWorks, supporting **2 wing types and 3 flap configurations** with continuously adjustable angle of attack and **0°-30° slat deflection in 5° steps**.
- **Engineered** the 3D CAD models for all wing and flap test geometries, translating 2D CFD-optimized airfoil sections into fabricable, wind-tunnel-ready hardware.

Group contributions:

- **Co-developed** the aircraft's powered-lift mathematical model using Kuhn's semi-empirical component build-up method, authoring **50% of the MATLAB implementation** and diagnosing a critical model-CFD correlation gap that redirected the team toward empirical k-factor calibration.
- **Achieved** a cruise **L/D of 22.417** and takeoff **L/D of 5.038** with a peak **C_L of 6.45**, validated through coupled mathematical modeling and 2D RANS CFD (Ansys Fluent, k- SST) -- placing **4th among 23 competing IITs** at Inter IIT Tech Meet 14.0, presented at IIT Patna.

PROJECT: Electricity generation from VAWT mounted on roof of automotive bluff body.

- Led a four-member team testing a roof-mounted, 3D-printed 150.82 mm Savonius rotor on a PLA automotive bluff body in a wind tunnel, **measuring drag by force balance, speed by pitot tube**, and selecting rotor type and placement through literature survey.
- Compared 3V and 5V generators across 5-23 m/s using multimeter measurements, showing the 3V unit starts near 5 m/s versus 13 m/s for the 5V **(attributing the later 5V start to higher torque)** and generates more power below their 19 m/s crossover.
- Evaluated blade width and pitch from -2° to 6° **across nine wind speeds from 6 to 23 m/s, with error bars**; the 6° pitch, 1.75×-width cup-blade rotor peaked at 422.37 mW at 23 m/s, 28.5% above the -2° case. Fitted linear and quadratic regressions to this data.

PROJECT : Autonomous VTOL-UAV Survey & Crop Analysis System

Tools Used: Python, OpenCV, YOLOv8, Shapely, Matplotlib

- **Designed an Energy-Efficient Drone Flight Planner:** Calculated the exact flight altitude 73 m and camera overlap needed to map a farm. Wrote a Python program to optimize the drone's flight path, reducing sharp turns by **25%** and saving **13.9% battery energy**.
- **Built an Automated Crop Counting Tool:** Developed an OpenCV Python script that isolates green crops from brown soil and separates overlapping plants, achieving **100% counting accuracy** in **0.16 seconds**.
- **Compared Traditional Image Processing vs. AI:** Fine-tuned a YOLOv8 AI model to detect crops and benchmarked it against OpenCV. Proved that traditional image processing was **29.6% more accurate** and **1.7x faster** for simple, structured crop patterns.

PROJECT : Hydrogen Fuel Cell Electric Generator Concept

(SolidWorks Aakruthi 2022 Competition | Team of 3)

Tools Used: SolidWorks (3D CAD & Assembly)

- **Designed a 130 kg Zero-Emission Generator Assembly:** Modeled a 646-component hydrogen fuel cell generator in SolidWorks to replace conventional, polluting diesel generators for remote off-grid power.
- **Integrated Subsystem Components & Operating Constraints:** Engineered the structural layout integrating the high-pressure hydrogen intake, fuel cell stack, compressor, and electrical wiring while accounting for ambient operating conditions .

PROJECT: Red Wine Quality Prediction | Python, pandas, seaborn, scikit-learn, Keras, PyTorch

- **Analysed** 1,599 red wine samples (11 chemical properties) with pandas, seaborn and matplotlib; **removed** 240 duplicates to prevent data leakage, and a correlation heatmap **identified** alcohol (+0.48) and volatile acidity (0.40) as the top quality drivers.
- **Built** Logistic Regression, Random Forest and Extra Trees classifiers in scikit-learn on a stratified 80/20 split, **lifting** accuracy from a 42.6% always-guess-5 baseline to 62.5% (+19.9 pts); alcohol ranked as the most important feature.

- **Implemented** the same neural network in Keras and PyTorch (hand-written training loop), **reaching 59.2% and 56.2%** accuracy; showed tree models outperformed it on this small dataset (1,087 training wines).

EXTRACURRICULAR

NSS Volunteer (7-Day Camp, Siddenahalli): Spearheaded eye-care medical camp setup, school painting, cleanliness drives, and tree plantation.

Achievements: Runner-Up in Intra-State Drama Competition, *"Explain the Physics Behind It"* Competition, and Spell Bee.

Interests: Sketching, Painting, Cycling, Cooking.